

Course Introduction

CUNY EES 79903

Seminar on Nature-Based Solutions

Brett Branco and Kyle McKay

September 1, 2026

Lecture Overview:

- Review of syllabus, schedule, and course project
- A brief history of nature-based solutions
- Tips on reading scientific papers and leading a discussion
 - Deep dive with Palmer and Febria (2012)
- Sign-ups for future discussions

Introductions

- Who are you?
 - Name, department, training, history,...
- What are your professional interests?
 - Career goals, field of study, etc.
- What brings you to this class?
 - My advisor made me, I want to learn about topic-A(!), Tuesdays are convenient for me,...
- What do you bring to this class?
 - Deep love of nature, disciplinary expertise, lived experience, dose of skepticism,...

Review of Syllabus

****Start with the boring details****

Key Items of Note

Logistics

- Time: Tuesdays 4:15-6:15pm
- Location: CUNY Grad Center (365 5th Ave)
 - In-person is FAR MORE INTERESTING for discussion-oriented classes.
- Instructors:
 - Kyle McKay, Ph.D., P.E., skmckay@gmail.com
 - Brett Branco, Ph.D., BBranco@brooklyn.cuny.edu
- Office hours on Mondays, Tuesdays or Thursdays at the CUNY grad center by appointment

Responsibilities

- Follow academic honesty policy
- Treat all class participants respectfully to create a safe, equitable, inclusive, supportive, and (ultimately) more productive environment that is free of harassment and discrimination, in which all feel welcome and valued.
- Read all assigned papers and participate actively in discussions.
- Jointly lead discussion on a minimum of 3 papers over the course of the semester.
- Class project

Grading

- Reading, preparation, and participation (50%)
- Leading discussion (25%)
- Project (25%)

Grades are defined as:

- A > 90%
- B = 80-89.99%
- C = 70-79.99%
- D = 60-69.99%
- F < 60%

All grades are final and assignments may not be revised or resubmitted.

A typical week

- Checking in and getting settled
- Updates from the “front lines” on NBS. Case studies, ongoing research, important happenings in the field, etc.
- Brief introductory lecture to the week’s content with an example NBS (~20 minutes)
- Discussion of papers (~20-30 minutes per paper)

Class Project

- As a class, we will compile a “state of the art” review of NBS
- Each student will compile 3-5 key messages (~1 page each) on 10 NBS-relevant topics
 - Key messages: the what, the so-what, the now-what
 - Traceable accounts: the evidence, the quality of the evidence, the level of agreement, key uncertainties
 - Topics listed in syllabus
- Review messages for two classmates
- Revise submission
- Pull all content into an ebook at the end of the semester

Two main texts are available for background reading (and lots of other materials will supplement)



Nature-Based Solutions for Cities

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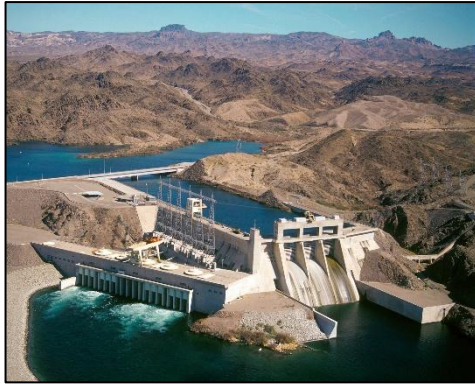
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A brief history of nature-based solutions



NETWORK FOR
ENGINEERING
WITH NATURE

What has been the global approach to infrastructure planning?



Resilient design

Expansion of project purposes

Mitigation hierarchy

Control of nature

Prior to 1970

1970s–1980s

1990s–2000s

2010s–2020s

Figures: Davis Dam (Reclamation), stream mitigation project (USACE), Cape Fear River fish bypass (Frank Yelverton), www.delawarelivingshorelines.org



NETWORK FOR
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WITH NATURE

Are infrastructure goals being met?

- Aging infrastructure
 - Increasing operations and maintenance
 - Rising fragility
 - Shifting societal needs
- New infrastructure
 - Increasing public safety
 - Improving quality of life
 - Growing populations
- Annual global spending on infrastructure (~\$2.7T, GI Hub 2021)

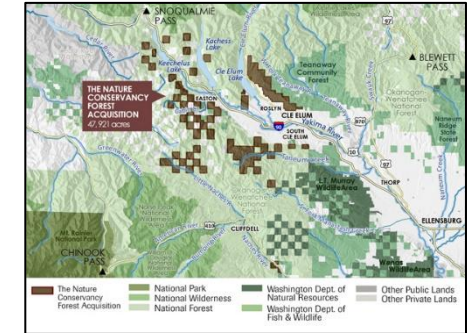
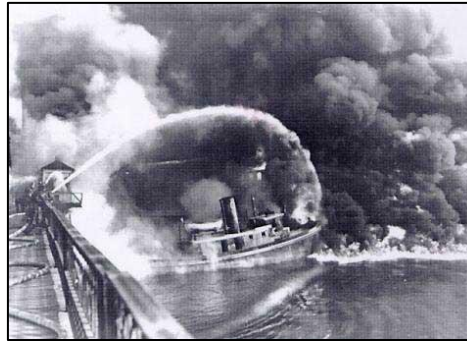


Figure: ASCE Infrastructure Report Card



NETWORK FOR
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What has been the global approach to conserving biodiversity (Mace 2014)?



Nature for itself

Nature despite people

Nature for people

People and
nature

Prior to 1970

1970s–1980s

1990s–2000s

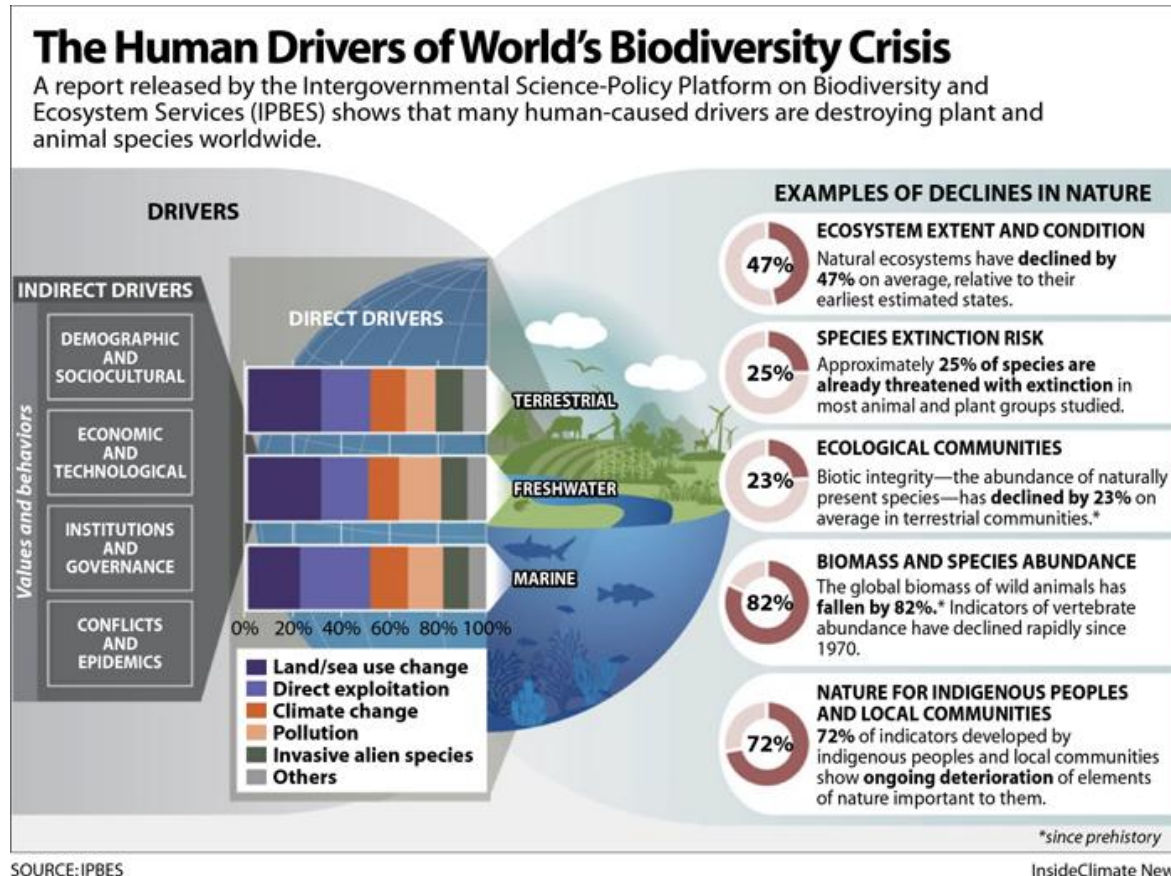
2010s–2020s

Figures: Yosemite National Park (NPS), Cuyahoga River fires (EPA), ecosystem services summary (Metro Vancouver), Washington's Eastern Cascade Mountain Range (TNC)



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Is biodiversity conservation working?



- None of the UN's 2020 biodiversity targets have been met (Obura et al. 2021)
- Extinctions occur at a pace orders of magnitude higher than background rates (IPBES 2019, Brauman et al. 2020)
- Annual spending on conservation would need to increase 6–7 fold to meet international goals (Deutz et al. 2020)



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Convergence of the historic arcs of these fields of practice

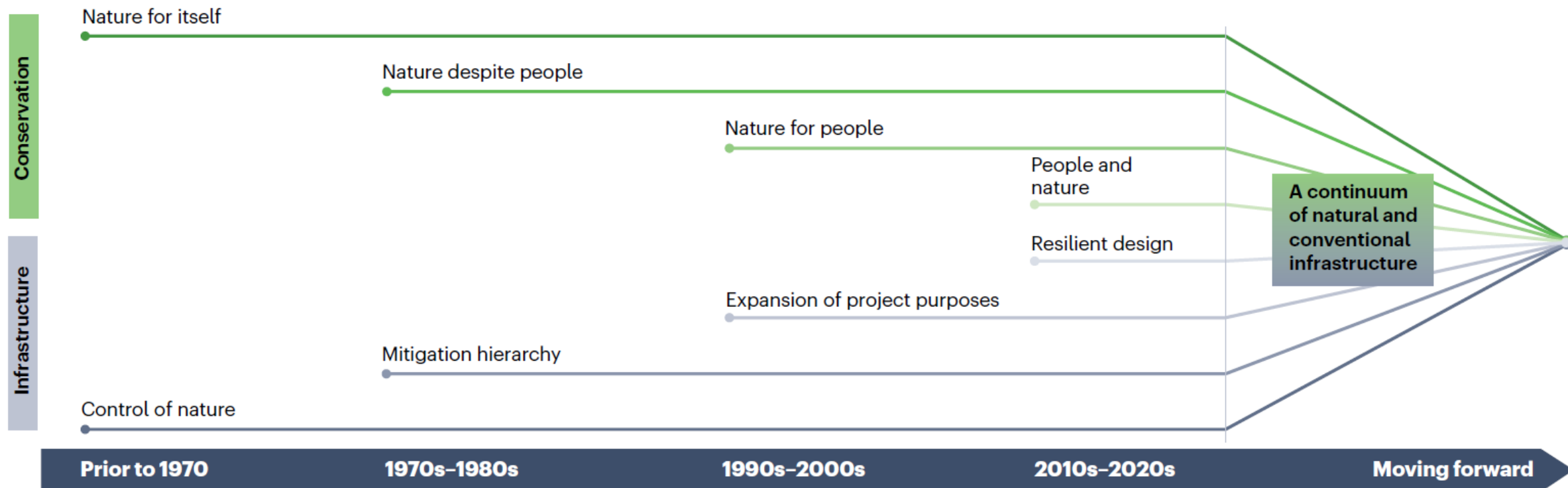


Figure: McKay et al. (2023, *Nature Reviews Earth & Environment*)



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A paradigm shift may be underway

A Conventional Framing of Infrastructure	An Emergent Approach to Infrastructure
Infrastructure performs engineering functions and services	Infrastructure performs engineering functions and services
Fewer objectives centered on an organization's mission	Many objectives with a multi-purpose framing
Independent funding => independent projects	Leveraged funding => systems outcomes
Ecosystems and social outcomes are treated as constraints or happy coincidences	Ecological and social benefits are included in the goals
Technical experts from a few disciplines develop alternatives	Technical experts lead alternative development, but other voices participate
Typically leads to more conventional, “grey” infrastructure	Merges conventional infrastructure with natural systems

Reading scientific literature and leading discussions

Other resources:

Drake J.M. 2026. Three rules for leading a paper discussion. BioScience , 2026, biag112
<https://doi.org/10.1093/biosci/biag112>.

Heard S.G. 2026. How to lead a journal club you won't be embarrassed by later. Scientist Sees Squirrel.
<https://scientistseessquirrel.wordpress.com/2026/01/13/how-to-lead-a-journal-club-you-wont-be-embarrassed-by-later/>.

What is a scientific “paper”?

- A. Tablet from the mount
- B. Scientific contribution
- C. Letter to colleagues
- D. Grant obligation
- E. Sisyphean exercise in futility
- F. A group project to come to consensus
- G. A condensation of other people’s work

Different types of papers and contributions

- Type of documents:
 - journals vs. government docs vs. “grey literature” vs. policy docs
- Types of journal papers:
 - forum/comment, perspective, primary contribution, review/synthesis
- Types of outlets:
 - disciplinary, broad readership, practice- vs. knowledge-oriented
- Types of writing teams:
 - single author vs. one lab vs. multi-institution synthesis

How do you decide what to read?

- What are your colleagues writing? What are they reading?
- Following people you respect
- Learning new approaches and methods
- Building context for your own work (historical and present)
- Keeping the pulse of the field
- How much should we be reading?
- How much should we be using AI?

Mechanics of reading (a deep dive into Palmer and Febria 2012)

ECOLOGY

The Heartbeat of Ecosystems

Which indicators of ecosystem structure and function must be measured to assess ecosystem health?

Margaret A. Palmer^{1,2,3} and Catherine M. Febria^{1,2}

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People generally equate health with a long and active life, but when it comes to ecosystems there is no agreement on what it means to be healthy. On page 1438 of this issue, Woodward *et al.* (1) show that different conclusions on the health of stream ecosystems can be reached depending on which combinations of metrics are used to assess them. The work points to an urgent need for a general framework for assessing ecosystem health.

An increasingly popular way to assess ecosystems is to evaluate the level of goods and services they provide to humans, such as water for drinking or irrigation (2). However, using ecosystem services as an indicator of ecosystem health is an oversimplification of the ecosystems services concept (3) and does not provide sufficient information on the extent to which those services can be provided sustainably (4–6). A second, older approach bases ecosystem health on its similarity to a least-impacted, reference, or historical state. This is the main approach in use today by water resource managers and typically involves measurements of what ecologists call ecosystem structure (7).

Ecosystem structure refers to attributes that can be evaluated with point-in-time measurements and that are assumed to reflect the existing status or condition of an ecosystem. Structural attributes may be easy to measure, but they do not capture the dynamic properties of an ecosystem that represent its actual performance. Functional measurements, on the other hand, attempt to capture system dynamics through repeated measurements that quantify key biophysical processes (see the table).

METRICS FOR EVALUATING THE HEALTH OF RIVER AND STREAM ECOSYSTEMS	
Structural metrics	Functional metrics
Biological diversity or species of interest	Productivity/reproduction, migration, trophic status
Native riparian vegetation width	Pollutant removal rates
Floodplain presence/width	Hydraulic retention
Canopy cover	Photosynthetic active radiation
Oxygen level	Biochemical oxygen demand, whole stream metabolism
Nitrogen, phosphorus concentrations	Nutrient cycling or flux rates
Pollutant concentrations	Pollutant removal or sequestration
Organic matter	Decomposition rate
Temperature	Thermal regime (magnitude, duration and timing)
Mean annual flow and depth	Flow regime (magnitude, duration and timing)
Turbidity	Sediment flux
Channel morphology	Channel migration, erosion rate
Streambed substrate	Streambed mobility

Ecosystem structure and function. The components of an ecosystem (structure) interact with dynamic biophysical processes (functions) to produce goods and services that people rely on from rivers and streams (e.g., fisheries, drinking water, irrigation, flood mitigation, recreation). Structural measurements (**left**) may be easier to make but provide less information than related functional measurements (**right**). In practice, these metrics vary widely. For example, nitrogen processing, commonly measured as denitrification potential, varies with organic matter by several orders of magnitude (21). Metabolism varies by two to three orders of magnitude (15, 17), as does decomposition rate depending on leaf type (1, 14). As Woodward *et al.* show, combinations of structural and functional measurements may be required to predict the state of an ecosystem or availability of a good or service.

Structure and function both contribute to ecosystem status and to the provision of goods and services, but what combination of measurements should be used to evaluate ecosystems is a matter of current scientific debate. Woodward *et al.* (1) evaluate river health on a continental scale using measurements or “health metrics”: the taxonomic composition of aquatic biota (structural), the eutrophication status based on nitrogen concentration (structural), and the decomposition rate of organic material in the form of leaf litter (functional). Their work suggests that management efforts to reduce nitrogen pollution may not lead to recovery of desired species or efficient processing of organic matter. The data show that the functional metric of decomposition can be quite variable even at some fixed level of water quality. Other factors known to influence the measurements they used, such as river geomorphology and land use, were not reported.

On the surface, the results might seem unsatisfying, but in fact they clearly demonstrate why it is important to use multiple metrics. Ecosystem components interact in complex ways, and ecological processes are context dependent (8); interpretation of measurements thus requires multiple types of data. For example, the rate of organic matter decomposition in rivers and streams is well-known to be influenced by the species and size of aquatic insects that tear apart decomposing leaves while feeding. Other key players that influence decomposition—microbes—increase their activity when nutrients are elevated and, like insects, also respond to other factors that typically change as rivers become eutrophic (9).

Recent studies have started to elucidate how these factors influence microbial production and community structure at large spatial and temporal scales (10). Such work is essential for understanding how to interpret and use metrics such as decomposition rate. Exploring patterns like those reported by Woodward *et al.* (1), coupled with clever natural and manipulative experiments (11, 12), is also an important step to take if ecologists hope to link suites of metrics to the desired future state of ecosystems. A review of more than 250 studies in the mid-Atlantic United States found that data are currently insufficient to identify which metrics will best reflect whether an ecosystem is on a clear trajectory toward improvement, because of the paucity of studies that couple functional measures with routine quantification of ecosystem structure (13).

Structural measures, such as the presence or absence of an iconic species like salmon, can indicate that something is wrong with an ecosystem, but typically not why. Thus, reliance on structure alone makes it difficult to prescribe ecosystem management or restoration actions. After all, doctors will routinely measure a sick patient's heart rate and blood pressure, rather than just their height, weight, and temperature. Human health metrics are based on long-term research that links metabolic processes to longevity and disease risk across highly diverse populations. Analogous studies for river and stream ecosystems are now accumulating (14–17).

Such an understanding could help to address several pressing questions: Which ecosystem functions should be the target of management efforts, given desired outcomes? Is temporal variation in ecological processes itself an important driver of ecosystem states, much like annual cycles of high and low flows are thought to influence species composition in rivers (18)? Which metrics can be used as early indicators of degradation or recovery, and how do they vary with each stressor? Can we predict the ecological consequences of managing for a specific ecosystem service?

The recently launched Intergovernmental Platform on Biodiversity and Ecosystem Services (19) and other large-scale efforts will have these questions to address. Given worldwide declines in species, and growing evidence that biodiversity may control and not just respond to ecological processes (20), more data such as those provided by Woodward *et al.* (1) on how ecosystem structure and function vary across a large range of conditions are urgently needed. Only then will ecologists have enough information to determine which set of metrics, under which

contexts, best equates to the human heart-beat for ecosystems.

References

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To move river science into the realm of effective health diagnosis before ecosystems collapse, researchers must work with natural resource managers to compile regionally specific data on how functional metrics vary over time and space—particularly in rivers considered to be least impacted. Knowledge of the range of natural variability, the factors that drive changes in ecosystem functions, and how these functions respond to management actions will generate a framework for sound prescriptions to improving ecosystem health.

Tips on reading scientific literature

- Skim a paper first by reading the abstract, glancing at the figures, and reading the conclusions. This is seeking to orient you to the basic flow, arguments, and rationale for the paper.
- Read the paper in detail.
- Explore topics that are unclear. Don't be afraid to look up terms that are unclear or read references cited.

Leading a discussion: Preparation matters!

- Reading the papers is rarely (if ever) enough to lead a group discussion.
- Thoughtful preparation and good notes will go a long way.
- Notes could take different formats:
 - A formal discussion guide for distribution to the class (e.g., one-page handout with overview and questions)
 - A short presentation or visual aids
 - Informal personal notes
- Don't be afraid to look up key citations, explore related topics online, or get help from others. Remember...you are leading us through the topic, so more preparation is needed!

Leading a discussion: Structure the conversation (Drake 2026)

- Start with a SHORT summary
 - The central finding
 - The method that produced it
 - The claim the authors build on it
 - Why any of it matters
- Plan for the life cycle of the discussion
 - Igniting the conversation with “closed” questions that build a shared baseline
 - Gaining momentum with “open” questions beyond repetition of the paper’s facts
 - Regulating the conversation with figures, tables, or structure
 - Close the conversation
- Ensure that disagreement is well-founded
 - What did the authors find?
 - How did the authors interpret it?
 - What do you think it means?

Leading a discussion:

Other prompts to consider

Some options that might be fun to think about. **See also Heard (2026).**

- What motivated the study? Who is involved? Who funded it? Why do we care? Is there a broader history we need to be aware of?
- Clarify any confusion over the methods or findings as a group. Are there any flaws that raise questions regarding the validity of the findings? Are there figures or tables that are particularly useful for summarizing the study?
- Discuss the broader application of the study's findings.
- Consider the philosophical underpinnings and assumptions of the analysis.
- Discuss the future applications of this study. Is it a shift in the wind, or simply more of the same story? What's next?
- Are there activities that will help clarify the topic or provoke further discussion? Feel free to reserve part of the time for an online tool, video, or infographic that you think drives home the point.

Leading a discussion: Mechanics

- You are in charge!
- Tangents are okay, but please bring us back to the point.
- Silence can be an effective tool for motivating people to speak. Don't be too quick to answer your own discussion questions.
- Don't let any one person drive the discussion. These can be challenging topics; we want to draw on everyone's experiences, not one individual's.

Participating in a discussion

- Be respectful and collegial.
- Help out the discussion leader! If you have compelling questions or topics, raise them. We are exploring the topic together.
- The blackboard and projector are available to all.
- Generally speaking, let three people share before you reengage.

Sign up for 3 dates!	Leader #1	Leader #2
Sep 8: NBS fundamentals	Grace	Jesse
Sep 15: NBS fundamentals		
Sep 22: Structured decision making and community engagement		
Sep 29: Analysis – ecological		
Oct 6: Analysis – co-benefits		
Oct 20: Analysis – engr performance		
Oct 27: Analysis – engr performance		
Nov 3: Alternatives analysis		
Nov 10: Performance-based design		
Nov 17: Constructability		
Nov 24: Performance monitoring		
Dec 1: Performance monitoring		
Dec 8: Life cycle analysis		
Dec 15: Course synthesis	McKay	Branco

Narrowing options for a field trip

- Days of the week?
- Periods out of town?